



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2024-2025
Course Handout (Part II)

Date: 31/07/2024

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course No. : CE F213
Course Title : SURVEYING
Instructor In-Charge : S. N. PATEL
Instructor : Sachin Solanki, Akash Samadhiya, Rakesh Kumar V G, Thakur Ramjiram Singh, Vijaykumar, Vivek Chahar

Course Description:

The compulsory disciplinary course has been designed to introduce the basic concepts of geodesy (surveying) for Civil Engineering students. Different basic and advanced methods of measurements and traversing have been included so that the student will be able to handle a given project independently, irrespective of whether it is a road or any other infrastructure project. Important issues like curve setting, calculation of areas and volumes, which form part and parcel of any filed civil engineer in his/her day to day activity. Hence in this course, these areas have also been included.

Scope & Objective:

The course introduces to the students, various basic techniques in surveying, viz. chain, compass, theodolite, plane table, tacheometry, traversing and calculation of areas and volumes along with fundamentals of a few advanced surveying techniques.

Text Books:

- T1. Duggal S.K.; Surveying; Tata McGraw-Hill, New Delhi, Vol I and II, 4th Edition (2013).
T2. Moondra, H.S. and Gupta Rajiv, Lab Manual for Civil Engineering, CBS, 2nd edition, (2000).

Reference Books:

- R1. Punmia B.C; Surveying; Laxmi Publishers, Vol I, II and III, (1990).
R2. Agor, R; A Text Book of Surveying & Levelling, Khanna Publishers. New Delhi.
R3. N N Basak: Surveying & Levelling, McGraw Hill Education (India) Private Limited, New Delhi, 2nd edition, 2014.
R4. Venkatramaiah, C; Textbook of Surveying, University Press, Second Edition, 2011.
R5. Bhavikatti, S.S; Surveying Theory & Practice, Ik International Publishing House Pvt Ltd, New Delhi, 201.

Course Plan:

Module	Lecture No.	Topic	Topics to be covered	Text Book
Module 1	1	Introduction to the basic concepts of Surveying	Fundamental definitions and concepts	Chapter 1/ T1 Vol I
	2-3	Linear Measurements and instruments	Methods, accessories, ranging	Chapter 2/ T1 Vol I
	4-6	Chain Survey	Steps in chain survey, field work and plotting in field book, obstacles in chaining.	Chapter 2/T1 Vol I



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Module 2	7-9	Compass Survey	Instrument, principles, Bearings	Chapter 3/TI Vol I
	20-22	Traversing	Methods, adjustments and plotting	Chapter 5/T1 Vol I
Module 3	10-11	Levelling	Instrument, Collimation method, Rise and fall method, curvature and refraction. Level book	Chapter 6/T1 Vol I
	12-13	Contouring	Objectives, use, methods.	Chapter 9/T1 Vol I
Module 4	14-16	Plane Table Survey	Accessories, methods, errors	Chapter 8/T1 Vol I
	17-19	Tacheometric Surveying	Theory, instrument constants, methods	Chapter 7/T1 Vol I
Module 5	23-27	Curve Ranging	Types, properties, circular and transition curves.	Chapter 11/T1 Vol I
Module 6	28-30	Trigonometrical Levelling	Geodetic survey, visibility between two places.	Chapter 1/T1 Vol II
Module 7	31-33	Computation of Areas	Different methods, approximate method, planimeter	Chapter 12/T1Vol I
	34-37	Computation of Volumes	Level section, multilevel section, volume from contour plan, mass-haul diagram	Chapter 13/T1 Vol I
Module 8	38-40	Advanced Topics	Total Stations and other advancements in surveying, Errors & adjustments, Photogrammetry and remote sensing	Chapter 10/T1Vol-I and Chapter 3, 5, 6,7,8,9/T1 Vol II
	41-42	Practical aspects of field work	Common mistakes in field, Setting out Works	Chapter 14/T1 Vol I

Module Learning objectives:

- To familiarize the basic fundamentals of surveying and linear measurements [**Module 1**]
- To compute and correct the measured bearings to determine area of field for different types of traverse [**Module 2**]
- To determine the reduced levels and also to establish various contours [**Module 3**]
- To plot the area of field using plane table surveying using the surveying principles and also the use of principle of tacheometry [**Module 4**]
- To set the simple circular curves for various geometric parameters [**Module 5**]
- To evaluate the errors and corrective measures due to the curvature, refraction and collimation [**Module 6**]
- To determine the areas and volumes for different road cross section and from contour maps [**Module 7**]
- To acknowledge the recent advancements and field innovations in land surveying [**Module 8**]



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Practicals:

No.	Name of experiment	Classes
1	Chain Surveying and Obstructions	2
2	Compass Traversing	1
3	Profile Levelling	1
4	Contour Survey by Square Grids	1
5	Practice session on Plane Table Surveying	1
6	Simple Circular Curve by Chain and Tape	1
7	Simple Circular Curve by Theodolite	1
8	Transition Curve by Theodolite	1
9	Ht. of Tall Objects by Two-Plane Method	1
10	Total Station Capabilities	1
11	Exploring Micro-Survey Software	1
12	Demonstration on GIS Software	1

Evaluation Scheme:

S.N o.	Evaluation Component	Duration	Weightage (%)	Date & Time	Nature of Component
1	Mid Sem	90 minutes	25	4.10.2024 (AN1)	Closed Book
2	Tutorials (Best 4 out of 6)	50 min	10 (=4x 2.5)		Open/Closed Book
3	Compre. Exam	3 hours	45	6.12.2024 (FN)	Closed Book
4	Lab Quiz	-----	10		Open/Closed Book
5	Lab Record (Hand Written)	-----	10		Open Book

Total marks for evaluation: 300 marks.

Chamber Consultation Hour: Thursday 4:00PM to 5:00PM.

Notices: Communications will be sent only through e-mail/Nalanda/google classroom.

Make-up Policy:

Make-up may be granted for genuine reasons only in Mid-Sem and Comprehensive examinations, upon request. There will be no make-up for Tutorials and Lab. Quizzes.

Instructor-in-charge
CE F213